REPORT

Experiment 1: Resistance Measurement

Measured tolerance equation = $\frac{|R_{ideal}-R_{measurement}|}{R_{ideal}} * 100\%$

	Color bands	Ideal resistance Value (without human body, Ω)	Ideal Tolerance (%)	Measured resistance Value (Ω)	Measured Tolerance (%)
Example: 100 Ohm Resistor	Brown Black Brown Gold	100	±5	98	2
1Ω Resistor	Brown Black Gold Gold	1	±5	1.04	4
1 kΩ Resistor	Brown Black Red Gold	1k	±5	1026	2.6
1 MΩ Resistor	Brown Black Green gold	1M	±5	0.988M	1.2
Human Body				0.5M	
1Ω Resistor & human body		0.999998	±5	0.999999	0.0001
1 kΩResistor & human body		985.056	±5	987	0.197
1 MΩ Resistor & human body		0.333M	±5	0.222M	33.3

Question:

What do you find in these data?

並聯會使總電阻變小，待測物並聯與之相比很大的電阻後，會使總體的相對誤差降低，若兩者之電阻值差不多（同個數量級），則誤差易較大

Does human body influence the resistance measurement? How do you explain it?

使總電阻變小，使總體的相對誤差降低（若並聯對象為遠小於他的電阻），人體的皮膚對電流有相

當大的阻抗，表示人體是相對較差的導體，人體的組成也會影響其電阻值。例如，肌肉和血液含有水分，因此它們具有較高的電導能力，而脂肪含水量較低，所以對電流的傳導能力較差。意味著肌肉部位的電阻值較低，而脂肪部位的電阻值較高。

Experiment 2: Set the DC Power Supply

Please set channel 1 and channel 2 at the same time. Use the multi-meter to measure the current and voltage apart. In the second mode, please set the +5V relative to GND and -5V relative to GND in the same time.

Mode		Channel 1		Channel 2	
		Voltage(V)	Current(A)	Voltage(V)	Current(A)
1. Independent	面板值	**5.0**	**0.5**	**3.3**	**0.3**
	測量值	4.9	0.51	3.2	0.31
2. Output +5V -5V and 0V	面板值	**5.0 (正電壓)**	**0.5**	**5.0 (負電壓)**	**0.5**
	測量值	5.1	0.51	-5.1	-0.51

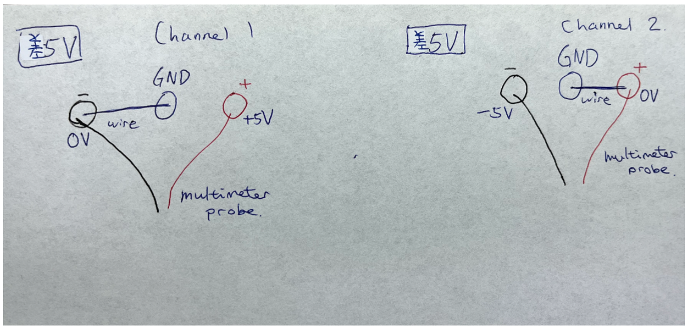
Draw the following connection: (for second mode)

Ground, positive and negative terminal

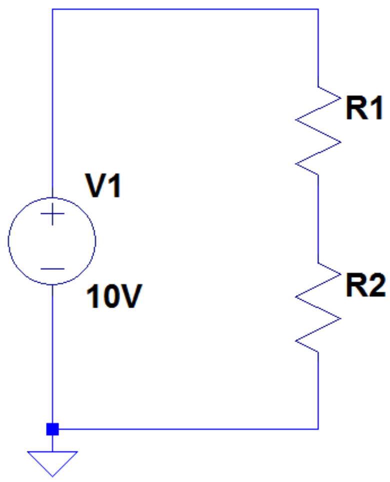
Multi-meter probe position

The Channel1's voltage is 5V. The Channel2's voltage is 5V. Please connect the cooper wire, set the GND point and make a +5V point and -5V point. Draw your result in the below figure and mark the voltage point.

因易跑版，故手繪



Experiment 3: The Effect Caused by the Internal Resistance of Multimeter



$$error = \frac{|V_1 - (V_{R1} + V_{R2})|}{V_1} \times 100\%$$

Condition	voltage（V_{R1}）	voltage（V_{R2}）	V_1	Error(%)
R_1=100Ω, R_2=100Ω	5.02	5.03	10.07	0.199
R_1=1MΩ, R_2=1MΩ	4.81	4.77	10.08	4.96

Question:

What do you find in these data?

所量測的電阻小時，誤差小

所量測的電阻大時，誤差大，內電阻產生的 Voltage Divider Effect 在高電阻時較顯著

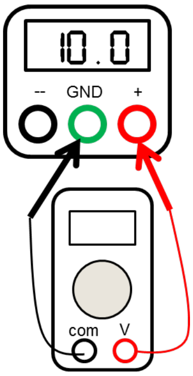
可能是所取之有效位數和內電阻之影響

What’s the influences of the internal resistance of multimeter?

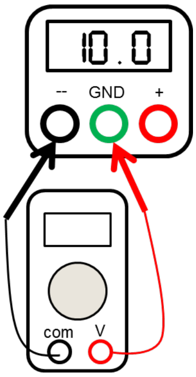
Voltage Divider Effect:內電阻和待測電阻串連，會使總電阻升高，流經電流下降，V=IR 則待測電阻測得之電壓差比實際較小。i.e.內電阻會造成待測電阻壓降。

另外，In high-resistance circuits, the internal resistance of the multimeter becomes a more significant portion of the total resistance in the circuit. For example, if you are measuring a 1MΩ resistor, a multimeter with an internal resistance of 1MΩ can introduce a 50% error, which is generally unacceptable for high-precision work. High-precision measurements often require very low measurement errors. In such cases, even a small error introduced by the internal resistance of the multimeter may exceed acceptable tolerances. In summary, while the internal resistance of a multimeter may have a relatively small impact on low-resistance circuits, it can become more significant in high-resistance circuits where precision and accuracy are paramount.

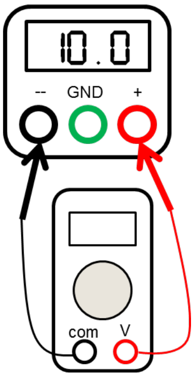
Experiment 4: Ground Concept Implement



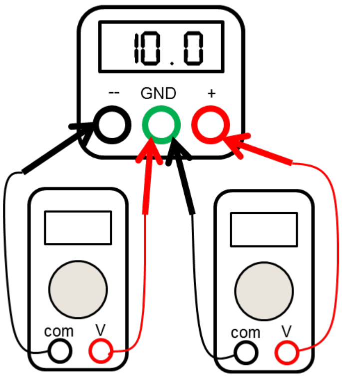
1. The voltage between “+” and “GND”： ___0___ V.



2. The voltage between “-” and “GND”： ___0___ V.



3. The voltage between “+” and “-”： ___10.1___ V.



4. Please connect two multi-meter in the same time.
The voltage between “+” and “GND”： ___5.1___ V,
and the voltage between “GND” and “-”： ___4.9___ V.